

Problem and contribution. Sequence-to-sequence models conventionally propagate state position by position with recurrent networks, making training within each example inherently sequential and placing long-range dependencies behind long computational paths. The paper introduces the *Transformer*, the first encoder-decoder transduction model whose sequence mixing uses only attention: every position can directly select information from every other position, while all positions are processed in parallel during training. The essential decomposition is *attention for cross-position communication* plus the same small feed-forward network independently at each position.

The information-routing primitive. Given queries Q , keys K , and values V ,

$$\text{Attn}(Q, K, V) = \text{softmax}\left(QK^T / \sqrt{d_k}\right) V.$$

Each query forms a content-dependent convex mixture of values. If query/key components are independent with unit variance, $\text{Var}(q \cdot k) = d_k$; division by $\sqrt{d_k}$ keeps logits near unit scale and avoids softmax saturation. One mixture can blur distinct relations, so h learned projections run this operation in parallel:

$$\begin{aligned} \text{MHA}(Q, K, V) &= \text{Concat}_{i=1}^h (H_i) W^O, \\ H_i &= \text{Attn}(QW_i^Q, KW_i^K, VW_i^V). \end{aligned}$$

Heads can select different positions and representation subspaces at roughly the cost of one full-width head because each uses reduced dimensions.

Architecture as a causal mechanism. The encoder maps all input tokens through six identical layers, each self-attention $\rightarrow$ position-wise FFN. The decoder uses six layers of masked self-attention $\rightarrow$ encoder-decoder attention $\rightarrow$ FFN: a triangular mask sets future-token logits to $-\infty$, preserving autoregressive factorization, while cross-attention queries the encoder output. Every sublayer is wrapped as $\text{LayerNorm}(x + \text{Sublayer}(x))$. The FFN is $\max(0, xW_1 + b_1)W_2 + b_2$, shared across positions but not layers. With no recurrence or convolution to encode order, sinusoidal position vectors are added to token embeddings; learned position embeddings performed almost identically in the paper’s ablation. The base model uses $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, $h = 8$, and $d_k = d_v = 64$; input embeddings, output embeddings, and the pre-softmax matrix share weights.

Why self-attention was computationally attractive.

For sequence length n , width d , convolution width k , and local-attention radius r , the paper compares:

Layer	work/layer	sequential steps	max path
Self-attention	$O(n^2d)$	$O(1)$	$O(1)$
Recurrent	$O(nd^2)$	$O(n)$	$O(n)$
Convolutional	$O(knd^2)$	$O(1)$	$O(\log_k n)$
Local attention	$O(rnd)$	$O(1)$	$O(n/r)$

Thus dense attention is cheaper than recurrence when $n < d$ —typical for the translation setup—and exposes constant-length dependency paths, but its quadratic sequence term is the central scaling weakness.

Evidence under the paper’s conditions. On WMT 2014 English–German, the base model reaches 27.3 BLEU at an estimated 3.3×10^{18} training FLOPs; the 213M-parameter “big” model reaches 28.4 BLEU, more than 2 BLEU above prior reported systems including ensembles, after 300k steps (3.5 days on eight P100 GPUs). For English–French, the table and abstract report 41.8 BLEU for the big model; the result prose says 41.0, an internal source inconsistency. Base training used 4.5M En–De pairs, Adam with 4000-step warmup then inverse-square-root decay, residual dropout 0.1, and label smoothing 0.1. Controlled En–De development ablations support multiple heads but not “more is always better”: one head scored 24.9 BLEU, 8/16 heads 25.8, and 32 heads 25.4. A lightly tuned four-layer Transformer also reached 91.3 F1 on WSJ-only constituency parsing and 92.7 semi-supervised, evidence of transfer beyond translation rather than proof of universal superiority.

Boundaries and interpretation. Dense attention trades training parallelism and short paths for $O(n^2)$ memory/work; the authors explicitly leave local attention for long inputs to future work. The decoder remains token-by-token at generation time, so recurrence is removed from the network layers, not from autoregressive decoding. Sinusoidal extrapolation beyond trained lengths is a hypothesis, not an established result. Attention plots show heads tracking long dependencies, anaphora, and syntactic-looking relations, but these examples are qualitative and do not establish that attention weights are faithful explanations. Results cover translation and parsing with substantial compute; architecture ablations are development-set experiments, not broad scaling laws.

Source: Vaswani et al., *Attention Is All You Need*, arXiv:1706.03762v7.